

22 Metric Spaces

23 Metric Spaces II



Mathematics
and Statistics

$$\int_M d\omega = \int_{\partial M} \omega$$

Mathematics 3A03 Real Analysis I

Instructor: David Earn

Lecture 22
Metric Spaces
Monday 10 March 2025

Announcements

- New, exciting topic today...

Metric Spaces

The metric structure of $\mathbb{R}$

Definition (Absolute Value function)

For any $x \in \mathbb{R}$,

$$|x| \stackrel{\text{def}}{=} \begin{cases} x & \text{if } x \geq 0, \\ -x & \text{if } x < 0. \end{cases}$$

Theorem (Properties of the Absolute Value function)

For all $x, y \in \mathbb{R}$:

- 1 $-|x| \leq x \leq |x|.$
- 2 $|xy| = |x| |y|.$
- 3 $|x + y| \leq |x| + |y|.$
- 4 $|x| - |y| \leq |x - y|.$

The metric structure of $\mathbb{R}$

Definition (Distance function or metric)

The distance between two real numbers x and y is

$$d(x, y) = |x - y| .$$

Theorem (Properties of distance function or metric)

- | | | |
|----------|----------------------------------|--|
| 1 | $d(x, y) \geq 0$ | <i>distances are positive or zero</i> |
| 2 | $d(x, y) = 0 \iff x = y$ | <i>distinct points have distance > 0</i> |
| 3 | $d(x, y) = d(y, x)$ | <i>distance is symmetric</i> |
| 4 | $d(x, y) \leq d(x, z) + d(z, y)$ | <i>the triangle inequality</i> |

Note: Any function satisfying these properties can be considered a “distance” or “metric”.

The metric structure of $\mathbb{R}$

Given $d(x, y) = |x - y|$, the properties of the distance function are equivalent to:

Theorem (Metric properties of the absolute value function)

For all $x, y \in \mathbb{R}$:

1 $|x| \geq 0$

2 $|x| = 0 \iff x = 0$

3 $|x| = |-x|$

4 $|x + y| \leq |x| + |y|$ (*the triangle inequality*)

Slick proof of the triangle inequality

Theorem (The Triangle Inequality for the standard metric on $\mathbb{R}$)

$|x + y| \leq |x| + |y|$ for all $x, y \in \mathbb{R}$.

Proof.

Let $s = \text{sign}(x + y)$. Then

$$|x + y| = s(x + y) = sx + sy \leq |x| + |y| ,$$

as required. □

A non-standard metric on $\mathbb{R}$

Example (finite distance between every pair of real numbers)

Let $f(x) = \frac{x}{1+x}$, and define $d(x, y) = f(|x - y|)$. Prove that $d(x, y)$ can be interpreted as a distance between x and y because it satisfies [all the properties of a metric](#).

Proof: The only metric property that is non-trivial to prove is the triangle inequality. Note that $f(x)$ is an increasing function on $[0, \infty)$, so the usual triangle inequality, $|x - y| \leq |x - z| + |z - y|$, implies

$$\begin{aligned} f(|x - y|) &\leq f(|x - z| + |z - y|) = \frac{|x - z| + |z - y|}{1 + |x - z| + |z - y|} \\ &= \frac{|x - z|}{1 + |x - z| + |z - y|} + \frac{|z - y|}{1 + |x - z| + |z - y|} \\ &\leq \frac{|x - z|}{1 + |x - z|} + \frac{|z - y|}{1 + |z - y|} = f(|x - z|) + f(|z - y|) \end{aligned}$$

i.e., $d(x, y) \leq d(x, z) + d(z, y)$. □

Poll

- Go to
https://www.childsmath.ca/childsa/forms/main_login.php
- Click on **Math 3A03**
- Click on **Take Class Poll**
- Fill in poll **Metric spaces: Is “ $=$ vs $\neq$ ” a metric?**
- .

Discrete metric

Example (Discrete metric on $\mathbb{R}$)

Let $d(x, y) = \begin{cases} 0 & x = y, \\ 1 & x \neq y. \end{cases}$ Is d a metric on $\mathbb{R}$?

By definition, $d(x, y)$ is non-negative, zero iff $x = y$, and symmetric. For the triangle inequality, if $x = y$ then $d(x, y) = 0$ so the inequality holds for any z . If $x \neq y$ then $d(x, y) = 1$, and at least one of x and y must not equal z , so the inequality says either $1 \leq 1$ or $1 \leq 2$.

Example (Discrete metric on any set X)

The argument that $d(x, y)$ is a metric on $\mathbb{R}$ has nothing to do with $\mathbb{R}$ specifically. $d(x, y)$ is a metric on any set X .

General metric space (X, d)

Definition (Metric space)

A **metric space** (X, d) is a non-empty set X together with a distance function (or **metric**) $d : X \times X \rightarrow \mathbb{R}$ satisfying

- 1 $d(x, y) \geq 0$ *distances are positive or zero*
- 2 $d(x, y) = 0 \iff x = y$ *distinct points have distance > 0*
- 3 $d(x, y) = d(y, x)$ *distance is symmetric*
- 4 $d(x, y) \leq d(x, z) + d(z, y)$ *the triangle inequality*

Much of our analysis of sequences of real numbers and topology of $\mathbb{R}$ generalizes to any metric space. Very often, definitions and proofs depend only on the existence of a metric, not on $|x|$ specifically. Many useful inferences can be made by identifying a metric on a space of interest.

Examples of metric spaces

Example (Metric spaces (X, d))

- $X = \mathbb{Q}$, with the standard metric $d(x, y) = |x - y|$.
As $\mathbb{Q} \subset \mathbb{R}$, each condition for d is satisfied in $\mathbb{Q}$.

How different is $(\mathbb{Q}, d)$ from $(\mathbb{R}, d)$?

- $X = \mathbb{N}$, with the standard metric $d(x, y) = |x - y|$.
As $\mathbb{N} \subset \mathbb{R}$, each condition for d is satisfied in $\mathbb{N}$.
- $X = \mathbb{R}^2$ with $d(x, y) = \sqrt{(x_1 - y_1)^2 + (x_2 - y_2)^2}$, where we write the vectors $x = (x_1, x_2)$, $y = (y_1, y_2) \in \mathbb{R}^2$.
- $X = \mathbb{R}^n$ with $d(x, y) = \sqrt{\sum_{i=1}^n (x_i - y_i)^2}$, where $x = (x_1, \dots, x_n)$, $y = (y_1, \dots, y_n) \in \mathbb{R}^n$.

This metric on $\mathbb{R}^n$ is called the *Euclidean distance*.

Metrics from norms

The Euclidean metric on $\mathbb{R}^n$ is the (Euclidean) length of the difference of two vectors. This connection between length and distance generalizes to any vector space in which *length* is defined.

Definition (Norm)

A **norm** on a vector space X is a real-valued function on X such that if $x, y \in X$ and $\alpha \in \mathbb{R}$ then

- 1 $\|x\| \geq 0$ and $\|x\| = 0$ iff x is the zero element in X ;
- 2 $\|\alpha x\| = |\alpha| \|x\|$;
- 3 $\|x + y\| \leq \|x\| + \|y\|$.

A vector space X equipped with a norm $\|\cdot\|$ is said to be a **normed vector space**. Any norm $\|\cdot\|$ **induces** a metric d via

$$d(x, y) = \|x - y\|.$$

Proving that a function is a norm is not necessarily easy. Let's try for the Euclidean norm. . . To that end, recall the notion of inner product . . .

Norms from inner products

Definition (Inner product)

An **inner product** on a vector space V over $\mathbb{R}$ is a function

$$\langle \cdot, \cdot \rangle : V \times V \rightarrow \mathbb{R}$$

such that for all $u, v, w \in V$ and all scalars $\alpha \in \mathbb{R}$:

- 1 $\langle u, v \rangle = \overline{\langle v, u \rangle}$ *conjugate symmetry*
- 2 $\langle \alpha u + v, w \rangle = \alpha \langle u, w \rangle + \langle v, w \rangle$ *linearity in 1st argument*
- 3 $\langle v, v \rangle \geq 0$ with equality iff $v = 0$ *positive definiteness*

A vector space equipped with an inner product is called an *inner product space*.

Definition (Inner Product Norm)

The norm induced by an inner product $\langle \cdot, \cdot \rangle$ is $\|u\| = \sqrt{\langle u, u \rangle}$.

Norms from inner products

Theorem (Cauchy-Schwarz inequality)

Let V be a (real) inner product space with inner product $\langle \cdot, \cdot \rangle$. For all vectors $u, v \in V$, we have

$$|\langle u, v \rangle| \leq \|u\| \|v\|$$

where $\|u\| = \sqrt{\langle u, u \rangle}$ is the norm induced by the inner product.

Proof.

The standard proof begins with an idea that probably took someone a long time to think of: Since $\langle v, v \rangle \geq 0$ for any $v \in V$, for any $t \in \mathbb{R}$ we have

$$\begin{aligned} 0 &\leq \langle u + tv, u + tv \rangle = \langle u, u \rangle + t \langle u, v \rangle + t \langle v, u \rangle + t^2 \langle v, v \rangle \\ &= \langle u, u \rangle + 2t \langle u, v \rangle + t^2 \langle v, v \rangle \end{aligned}$$

This is a quadratic polynomial in t , which is non-negative for all $t \in \mathbb{R}$. Hence, this quadratic has at most one real root. Consequently, its discriminant is non-positive, i.e., $(2 \langle u, v \rangle)^2 - 4 \langle u, u \rangle \langle v, v \rangle \leq 0$.

continued...

Norms from inner products

Proof of Cauchy-Schwarz inequality (continued).

Simplifying the non-positive discriminant condition, we have

$$(\langle u, v \rangle)^2 \leq \langle u, u \rangle \langle v, v \rangle .$$

Taking square roots, we have

$$|\langle u, v \rangle| \leq \|u\| \|v\| ,$$

as required. □

How might you come up with such a proof?

Perhaps by guessing the result (based on knowing it in $\mathbb{R}^2$) and then working backwards.



Mathematics
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$$\int_M d\omega = \int_{\partial M} \omega$$

Mathematics 3A03 Real Analysis I

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Lecture 23
Metric Spaces II
Wednesday 12 March 2025

Announcements

- Participation deadline for Assignment 4 was at 11:25am today.

Last time...

- Introduction to **metric spaces**
 - Critical ingredient: the triangle inequality
- **Cauchy-Schwarz inequality**
(proved for real inner product spaces)

Norms from inner products

If X is an inner product space, then Cauchy-Schwarz allows us to prove that the induced norm really is a norm (i.e., satisfies the triangle inequality). For $x, y \in X$, we have

$$\begin{aligned}\|x + y\|^2 &= \langle x + y, x + y \rangle \\ &= \langle x, x \rangle + \langle x, y \rangle + \langle y, x \rangle + \langle y, y \rangle \\ &= \|x\|^2 + \|y\|^2 + 2 \langle x, y \rangle \\ &\leq \|x\|^2 + \|y\|^2 + 2 \|x\| \|y\| \\ &= (\|x\| + \|y\|)^2\end{aligned}$$

$$\implies \|x + y\| \leq \|x\| + \|y\| .$$

In particular, $\mathbb{R}^n$ with the usual “dot product” $\langle x, y \rangle = \sum_{i=1}^n x_i y_i$ induces the Euclidean norm $\|\cdot\|_2$, which therefore really is a norm, and $d(x, y) = \|x - y\|_2$ really is a metric (the Euclidean distance).

What about other norms induced by inner products?

Other metric spaces induced by inner products

We are accustomed to finite vectors:

$$x = (x_1, x_2) \in X = \mathbb{R}^2$$

$$x = (x_1, x_2, x_3) \in X = \mathbb{R}^3$$

$$x = (x_1, x_2, \dots, x_n) \in X = \mathbb{R}^n$$

We can think of a sequence as an infinite vector:

$$x = (x_1, x_2, \dots) \in X = \{\{x_n\} : n \in \mathbb{N}\}$$

The points in this space (X) are infinite-dimensional vectors.

We can think of an infinite vector as a function:

$$x_n = f(n) \quad \implies \quad (x_1, x_2, \dots) = (f(1), f(2), \dots)$$

The points in this space are functions: $X = \{f \mid f : \mathbb{N} \rightarrow \mathbb{R}\}$.

So we can generalize to other spaces via functions, e.g.,

$$C[0, 1] = \{f : [0, 1] \rightarrow \mathbb{R} \mid f \text{ continuous}\}$$

All of the above spaces have a natural inner product, and hence a natural norm and metric.

Other metric spaces induced by inner products

Inner products that convert the spaces on the previous slide into (Euclidean) metric spaces:

$$\mathbb{R}^n: \quad \langle x, y \rangle = \sum_{i=1}^n x_i y_i$$

$$\ell^2(\mathbb{R}): \quad \langle x, y \rangle = \sum_{i=1}^{\infty} x_i y_i$$

$$C[a, b]: \quad \langle f, g \rangle = \int_a^b f(x)g(x) dx$$

Note: ℓ^2 includes only **square-summable** sequences: $\sum_{n=1}^{\infty} x_n^2 < \infty$

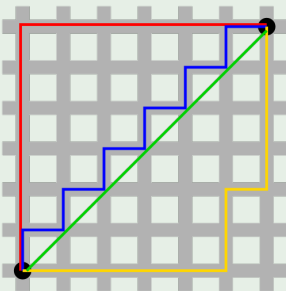
*Do we need to specify that $C[a, b]$ contains only **square-integrable** functions?*

Metrics from norms

Example (Taxicab distance)

Taxicab norm on $\mathbb{R}^n$

$$\|x\|_1 = \sum_{i=1}^n |x_i|$$



In taxicab geometry, the lengths of the **red**, **blue**, **green**, and **yellow** paths all equal 12, the taxicab distance between the opposite corners, and all four paths are shortest paths. Instead, in Euclidean geometry, the **red**, **blue**, and **yellow** paths still have length 12 but the **green** path is the unique shortest path, with length equal to the Euclidean distance between the opposite corners, $6\sqrt{2} \approx 8.49$.

Image and caption from [Wikipedia article on "Taxicab geometry"](#).

Metrics from norms

Example (p -metric)

p -norm on $\mathbb{R}^n$ (for $p \geq 1$) $\|x\|_p = \left(\sum_{i=1}^n |x_i|^p \right)^{\frac{1}{p}}.$

- $p = 1$ is the taxicab norm.
- $p = 2$ is the Euclidean norm.

What happens as $p \rightarrow \infty$? For any $x \in \mathbb{R}^n$, we have

$$\begin{aligned} \|x\|_p &= \left(\sum_{i=1}^n |x_i|^p \right)^{\frac{1}{p}} = |x_k| \left(\sum_{i=1}^n \left| \frac{x_i}{x_k} \right|^p \right)^{\frac{1}{p}} \quad (|x_k| > |x_i| \quad \forall i \neq k) \\ &= |x_k| \left(1 + \sum_{i \neq k} \left| \frac{x_i}{x_k} \right|^p \right)^{\frac{1}{p}} \xrightarrow{p \rightarrow \infty} |x_k| \end{aligned}$$

What further work is required if $\nexists k$ s.t. $|x_k| > |x_i| \quad \forall i \neq k$?

Therefore, we define $\|\cdot\|_\infty$ to be

Max norm on $\mathbb{R}^n$ $\|x\|_\infty = \max_{1 \leq i \leq n} |x_i|$

Poll

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- .

Metrics from norms

Example (p -metric)

Proving that $p = 1$ and $p = \infty$ yield norms is a *good exercise*.

Only $p = 2$ is induced by an inner product.

That the p -norms for $p \neq 1, 2, \infty$ are norms is harder to prove (but true), so

$$d_p(x, y) = \|x - y\|_p$$

is a metric on $\mathbb{R}^n$ for any $p \geq 1$.

Topology of metric spaces

We can generalize the notion of “neighbourhood of a point” to any metric space:

Definition (Open ball)

Let (X, d) be a metric space. If $x_0 \in X$ and $r > 0$ then the **open ball of radius r about x_0** is

$$B_r(x_0) = \{x \in X : d(x, x_0) < r\}.$$

x_0 is said to be the **centre** of $B_r(x_0)$.

Note: The notation $B(x_0, r)$ is also common (and used in TBB).

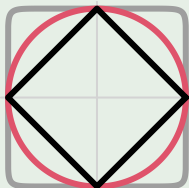
Definition (Neighbourhood)

A **neighborhood** of x is any set that contains an open ball $B_r(x)$ for some $r > 0$.

Topology of metric spaces

Example (Open balls in metric spaces)

- In the metric space $(\mathbb{R}, \text{standard})$, i.e., $\mathbb{R}$ with $d(x, y) = |x - y|$, $B_r(x) = (x - r, x + r)$, an open interval of length $2r$ centred at x .
- In $\mathbb{R}^n$ with Euclidean metric $d(x, y) = \|x - y\|_2$, $B_r(x)$ has a spherical boundary (circular boundary if $n = 2$).
- In $\mathbb{R}^n$ with a p -norm $\|\cdot\|_p$, the ball is not spherical. For $\mathbb{R}^2$ with the Taxicab metric $d(x, y) = \|x - y\|_1$, $B_r(x)$ is diamond shaped, and for $\mathbb{R}^2$ with the Max norm $\|\cdot\|_\infty$, $B_r(x)$ is a square. We write $B_r^p(x)$ for balls in the p -norm (" **p -balls**").



$$B_r^p(x)$$

- $p = 1$
- $p = 2$
- $p = 16$

Topology of metric spaces

Example (p -norm inequalities and containments)

The following inequalities relate the various p -norms on $\mathbb{R}^n$,

$$\|x\|_{\infty} \leq \|x\|_2 \leq \|x\|_1 \leq n \|x\|_{\infty}, \quad \text{and} \quad \|x\|_1 \leq \sqrt{n} \|x\|_2.$$

A good exercise.

Balls in the norm $\|\cdot\|_p$ are often written $B_r^p(x)$. The inequalities above imply that the following sets are *nested*:

$$B_{r/n}^2(x) \subset B_{r/n}^{\infty}(x) \subset B_r^1(x) \subset B_r^2(x) \subset B_r^{\infty}(x).$$

Another good exercise.

Example (Balls in the discrete metric)

For any set X , in the *discrete metric* the balls are simple, but strange. If $0 < r \leq 1$ then $B_r(x) = \{x\}$, a single point! If $r > 1$ then $B_r(x) = X$, the whole space! You can't be "close" to a point x unless you are at x itself!

Topology of metric spaces

Definition (Convergence of a sequence in a metric space)

Let (X, d) be any **metric space**. A sequence $(x_n)_{n \in \mathbb{N}}$ **converges** to $x \in X$, written $x_n \xrightarrow{n \rightarrow \infty} x$ or $\lim_{n \rightarrow \infty} x_n = x$, if:

$$\forall \varepsilon > 0, \exists N \in \mathbb{N} \text{ } \} \text{ } d(x_n, x) < \varepsilon \quad \forall n \geq N.$$

If the sequence does not converge to any $x \in X$, we say it **diverges**.

Equivalently, $x_n \xrightarrow{n \rightarrow \infty} x$ if $\forall \varepsilon > 0, \exists N \in \mathbb{N}$ with

$$x_n \in B_\varepsilon(x) \quad \forall n \geq N,$$

is, for any neighborhood $B_\varepsilon(x)$ of x , x_n lies inside that neighborhood eventually, and stays inside that neighbourhood.

Definition (Boundedness in a metric space)

In any **metric space** (X, d) , a sequence (x_n) is **bounded** if there exists $x_0 \in X$ and $r > 0$ such that $x_n \in B_r(x_0)$ for all $n \in \mathbb{N}$.

Theorem

In any metric space (X, d) , any convergent sequence is bounded.

Note: The converse is FALSE. e.g., $x_n = (-1)^n$ in $(\mathbb{R}, \text{standard})$.

Proof.

Taking $\varepsilon = 1$ (or any particular value) in the definition of convergence, since $x_n \xrightarrow{n \rightarrow \infty} x$ in (X, d) , $\exists N \in \mathbb{N}$ with $d(x_n, x) < \varepsilon = 1$, $\forall n \geq N$, i.e., $x_n \in B_1(x)$, $\forall n \geq N$. The earlier elements of the sequence, $x_1, \dots, x_{N-1}$, are a finite collection, so we can choose $r > 0$ so that

$$r > \max\{d(x_1, x), d(x_2, x), \dots, d(x_{N-1}, x), 1\}.$$

With this r , $x_n \in B_r(x)$ holds $\forall n \in \mathbb{N}$. □